Solution Challenge 2026

Digital Asset Protection System

An AI-powered multi-agent platform for sports media rights protection using Google ADK · Gemini 2.5 Flash · Vertex AI · Cloud Run

Team Details

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c. Problem Statement

Sports media rights are routinely violated through unauthorized streaming, piracy, and redistribution. Rights holders lack a unified, intelligent system to register, monitor, flag, and report IP violations in real time — resulting in massive revenue loss and unenforceable licensing.

Brief About the Solution

Digital Asset Protection System

An end-to-end AI agent system built on Google ADK that enables sports media rights holders to register official assets, detect IP violations, flag unauthorized usage, and generate protection reports — all through natural language conversation.

The system uses a primary coordinator agent that routes requests to 3 specialist sub-agents, each backed by purpose-built Python tools. Deployed live on Google Cloud Run with Gemini 2.5 Flash as the reasoning engine.

Live Deployed

my-agents-807379743631
.us-central1.run.app

4+ AI Agents

Multi-agent ADK architecture
with Gemini 2.5 Flash

6 Tools

register, list, flag, search,
report, update_status

Stack: Google ADK · Gemini 2.5 Flash · Vertex AI · MCP Toolbox · Cloud Run · Python · Docker

Opportunities

a. Differentiation

- Unlike static rights management tools, our system uses a conversational AI agent — users simply describe violations in plain English
- Multi-agent specialization: each sub-agent is an expert in its domain (registration, violation detection, reporting)
- Live Cloud Run deployment with real ADK Web UI — not a prototype, a working production system

b. How It Solves the Problem

- Rights holders register assets and get instant violation alerts — no manual monitoring needed
- Gemini 2.5 Flash reasons over all 6 tools to give accurate, context-aware responses
- Reports give a full picture of open violations, critical alerts, and license status in seconds

c. USP

- Only solution combining Google ADK multi-agent architecture + MCP Toolbox + Vertex AI for sports IP protection
- Natural language interface — no technical expertise required from rights holders
- Publicly accessible live demo at my-agents-807379743631.us-central1.run.app

List of Features Offered by the Solution

1

Asset Registration

Register sports media assets with owner, type, license expiry. Duplicate detection built-in.

2

Asset Listing

List all registered assets with live status: active, archived, pending_review, suspended.

3

Violation Flagging

Flag unauthorized usage with URL, description, and severity (low/medium/high/critical).

4

Violation Search

Search violations by asset ID, keyword, owner name, URL, or severity level.

5

Protection Reports

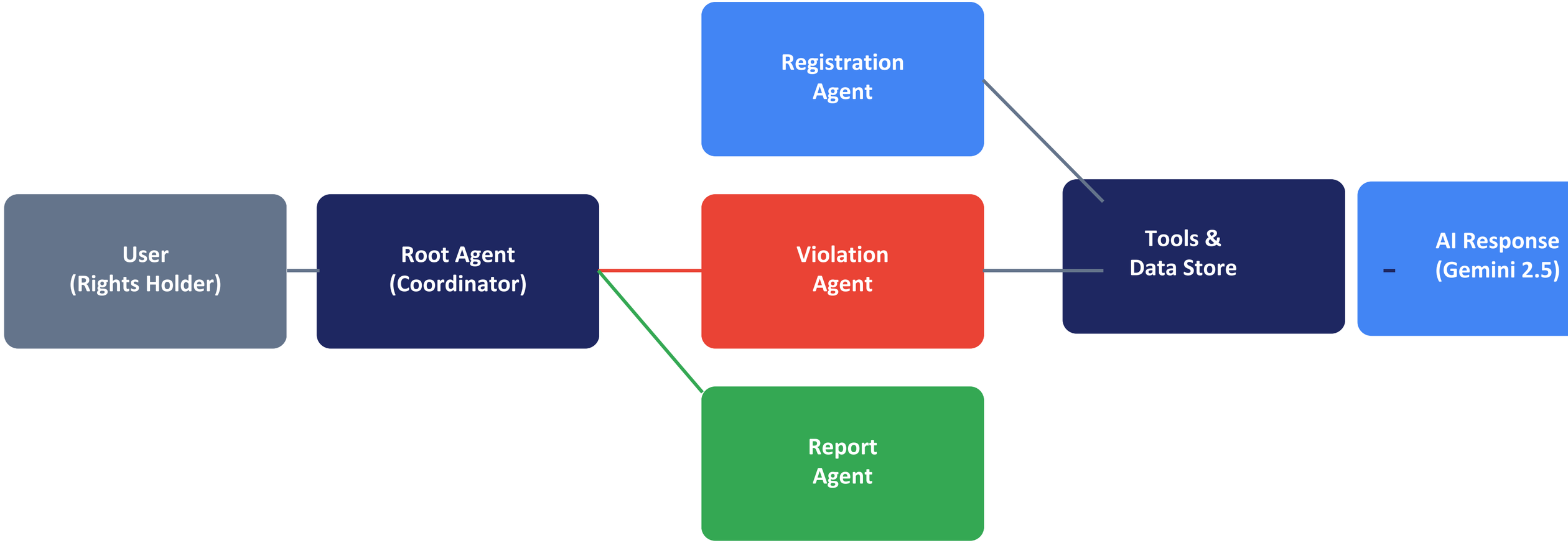
Generate per-asset or full portfolio reports with violation counts and severity breakdown.

6

Status Management

Update asset lifecycle status; graceful degradation if toolbox is unavailable.

Process Flow Diagram



6 MCP Tools:

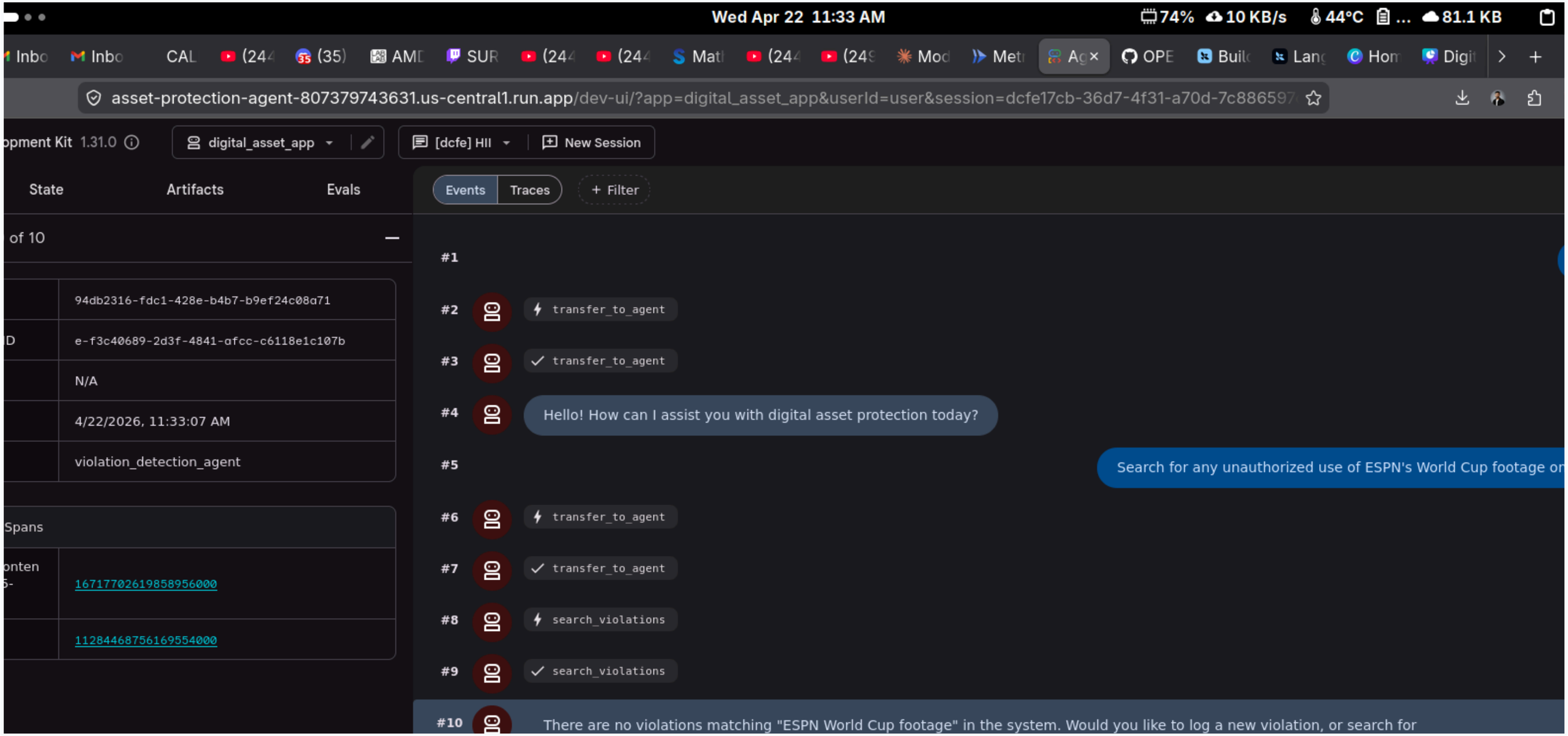
register_asset

get_assets_by_owner

get_violations

get_report_status

Wireframes / Mock UI — ADK Web Developer Interface



Architecture Diagram

User / Browser

Cloud Run — my-agents (ADK Web Server + Python Agent)

Root Agent (Gemini 2.5 Flash — Coordinator)

Registration Agent	Violation Agent	Report Agent
register_asset list_assets update_status	flag_violation search_violations	get_asset_report

Infrastructure Layer:

- Vertex AI / Gemini API
- MCP Toolbox SQL Tools
- Google Cloud Run
- Docker Container

Technologies Used in the Solution

AI/Agent	AI Model	Cloud AI	Tools
Google ADK Multi-agent orchestration framework	**Gemini 2.5 Flash** LLM for reasoning & tool selection	**Vertex AI** Managed AI platform & auth	**MCP Toolbox** SQL tools exposed as agent tools

Cloud	Language	DevOps	Data
Google Cloud Run Serverless container deployment	**Python 3.12** Agent code & tool functions	**Docker** Containerized build & deploy	**BigQuery** Public datasets & data queries

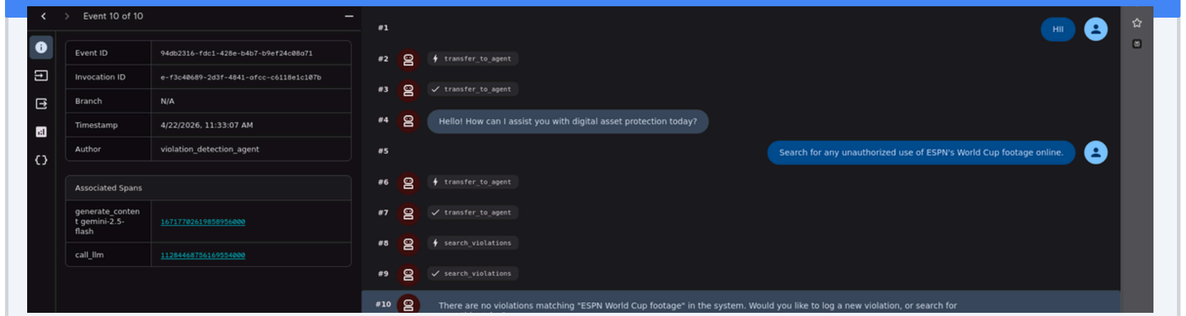
Estimated Implementation Cost (Prototype Phase)

Google Cloud Run Min: 0, Max: 3 instances · Auto-scaling	**$0–15/mo**
Gemini 2.5 Flash Input: $0.075/M tokens · Output: $0.30/M	**~$5–20/mo**
Vertex AI Auth Service account — included in GCP free tier	**$0**
MCP Toolbox Self-hosted container on Cloud Run	**$0–5/mo**
BigQuery Public datasets — first 1TB free/month	**$0**
Cloud Build First 120 build-minutes/day free	**$0**

Total estimated prototype cost: $5–40/month | Current hackathon usage: ~$0 (GCP free credits)

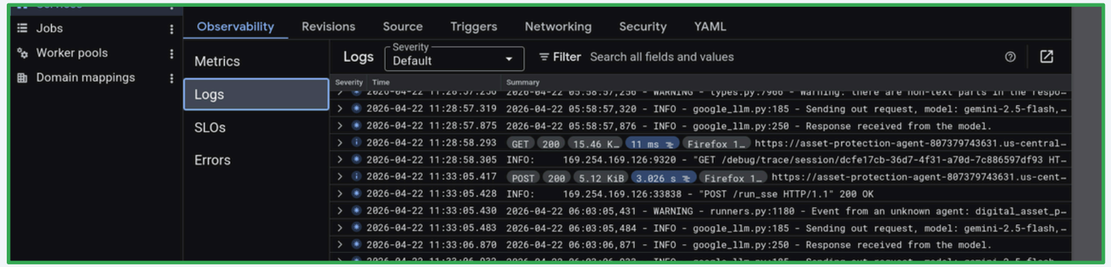
Snapshots of the MVP

ADK Web UI — Live Agent



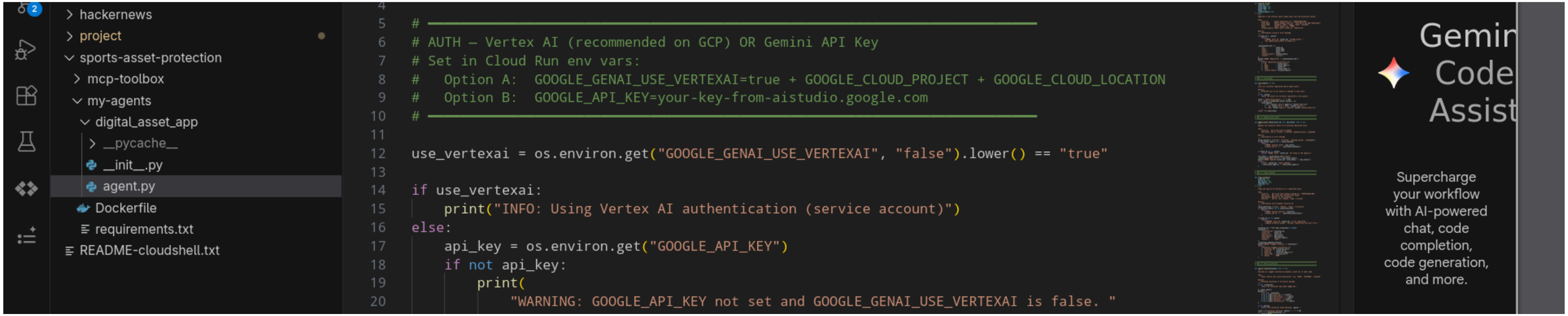
digital_asset_app running on Cloud Run with session management, Events & Traces tabs

Cloud Run Logs



Successful tool loading, session creation, HTTP 200 OK responses for all endpoints

Agent Code — agent.py



Additional Details / Future Development

Phase 1
DB

BigQuery Integration

Migrate in-memory store to BigQuery — persist all assets and violations across sessions with full query capability.

Phase 2
AI

Automated Web Scanning

Add a web-scan sub-agent that proactively searches for unauthorized usage using Google Search API + Gemini vision.

Phase 3
UI

Rights Holder Dashboard

Build a React/Next.js web dashboard showing violation heatmaps, asset health scores, and takedown request status.

Phase 4
API

API & Webhook Alerts

Expose REST API for external integrations + email/Slack webhook alerts when critical violations are detected.

Target: Expand from sports IP to general digital media (music, film, publishing) with multi-industry support

Project Links

1

GitHub Public Repository

https://github.com/OPENSOURCE6012/digital-asset-protection/tree/main

Working Prototype Link

https://asset-protection-agent-807379743631.us-central1.run.app

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Thank You!